Grade 8 Exam Paper November
ATP Based on Term 3 and Term 4 Work

Name: _____ Date: _____

Class: _____ Total: _____ / 100

Time: 2 Hours

Instructions

1. Answer ALL the questions.
2. Write neatly and legibly
3. Use a ruler where necessary.
4. Read each question carefully.

Question 1 (12 marks)

1. Simplify the following expressions:

a) $3x(4x + 5) - 4x(7x - 8)$ (3)

b) $\frac{27x^3 + 33x^2 - 12x}{3x} - 4x(5x - 8)$ (4)

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c) $\sqrt{(81x^4)} + (-4x)^2$ (3)

2. Find the value of the following expression if a = -1, b = 3, and c = 2 (2)

$ab^2 + bc - abc$

(12)

Question 2 (14 marks)

1. Complete this table of ordered pairs for the equation $y = 3x - 4$ (5)

x	-3	0	3	5	
y					20

2. Find the value of x in each of these equations:

a) $7x + 2 = 23$ (2)

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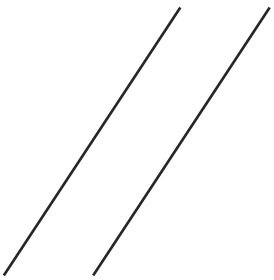
b) $3x^2 - 5 = 70$ (3)

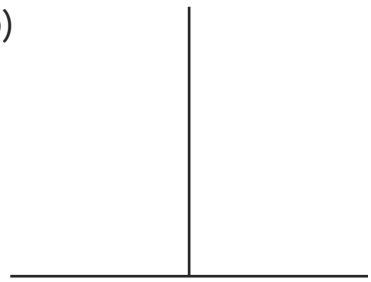
c) $\frac{x}{3} + 4 = 2(x - 7)$ (4)

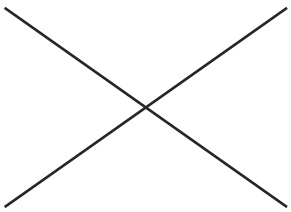
(14)

Question 3 (14 marks)

1. Name each of these types of lines: (3)

a) 

b) 

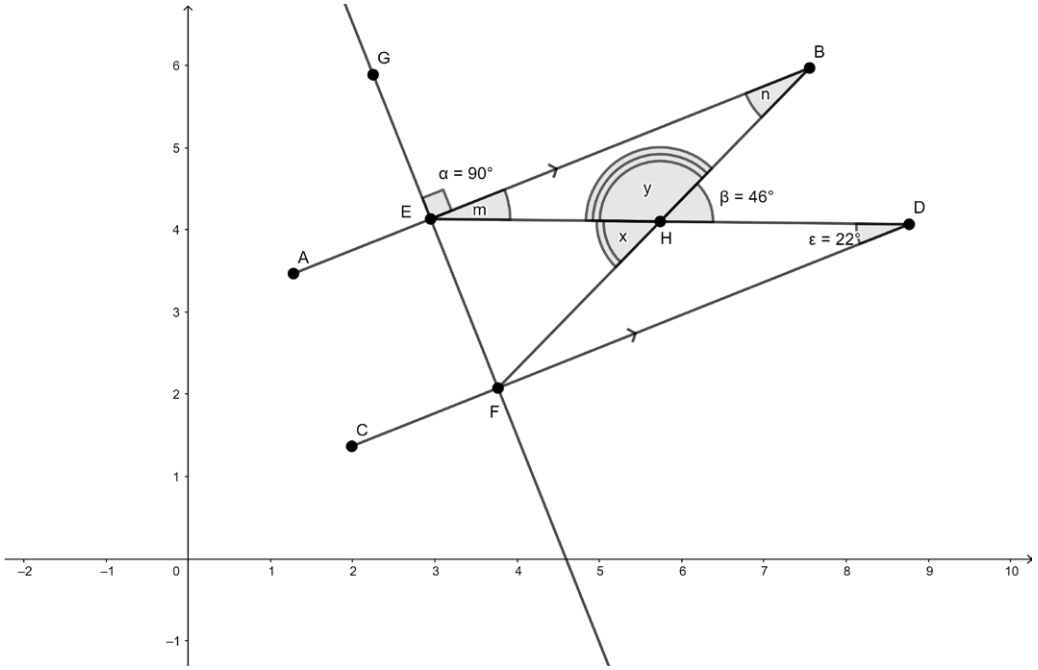
c) 

2. Solve for the variables x, y, m and n in this diagram:

   In the diagram below,

   AB || CD, and GF ⊥ AB, $H\hat{D}F = 22°$, and $B\hat{H}D = 46°$

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Statement	Reason

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Question 4 (19 marks)

1. Match the shape in column A with the description given in column B. (7)

Shape	Description
1. Parallelogram	a. A quadrilateral where the two sets of adjacent sides are equal.
2. Equilateral triangle	b. A triangle where two sides are equal.
3. Kite	c. A quadrilateral with two pairs opposite sides equal and parallel with right angles.
4. Isosceles triangle	d. A quadrilateral with one set of sides parallel.
5. Rhombus	e. A quadrilateral with two pairs of opposite sides parallel.
6. Rectangle	f. A quadrilateral with all four sides equal.
7. Trapezium	g. A triangle with all 3 sides equal.

1._____2._____3._____4._____5._____6._____7._____

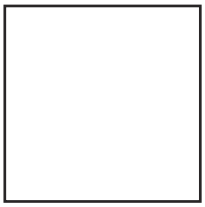
2. How do we know when two shapes are congruent? (2)

_____

_____

_____

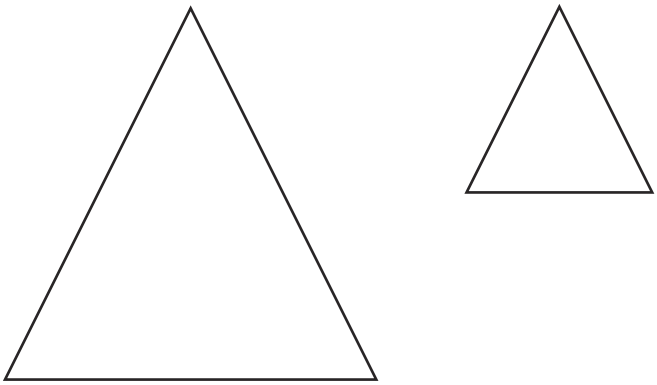
3. State whether each pair of shapes is similar, congruent, or neither. (3)

a)  

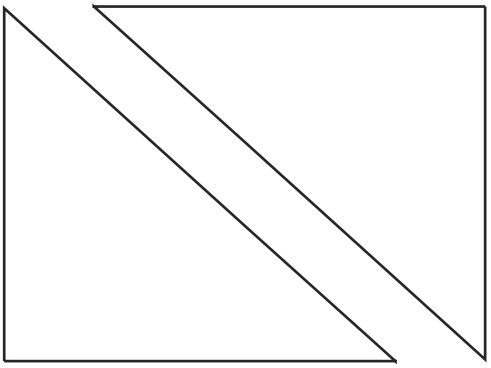
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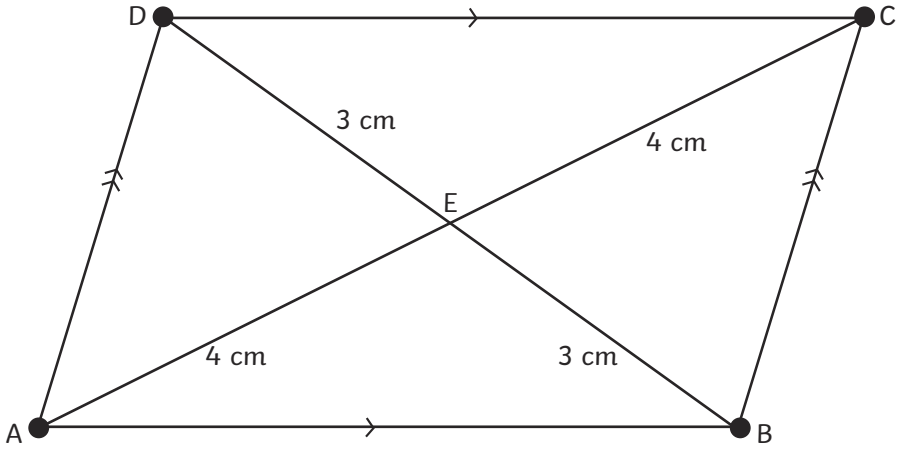
b)



c)



4. In the diagram, AB || DC and AD || BC.
We are given that DE = EB = 3cm and AE = EC = 4cm.



a) Name three sets of congruent triangles: (3)

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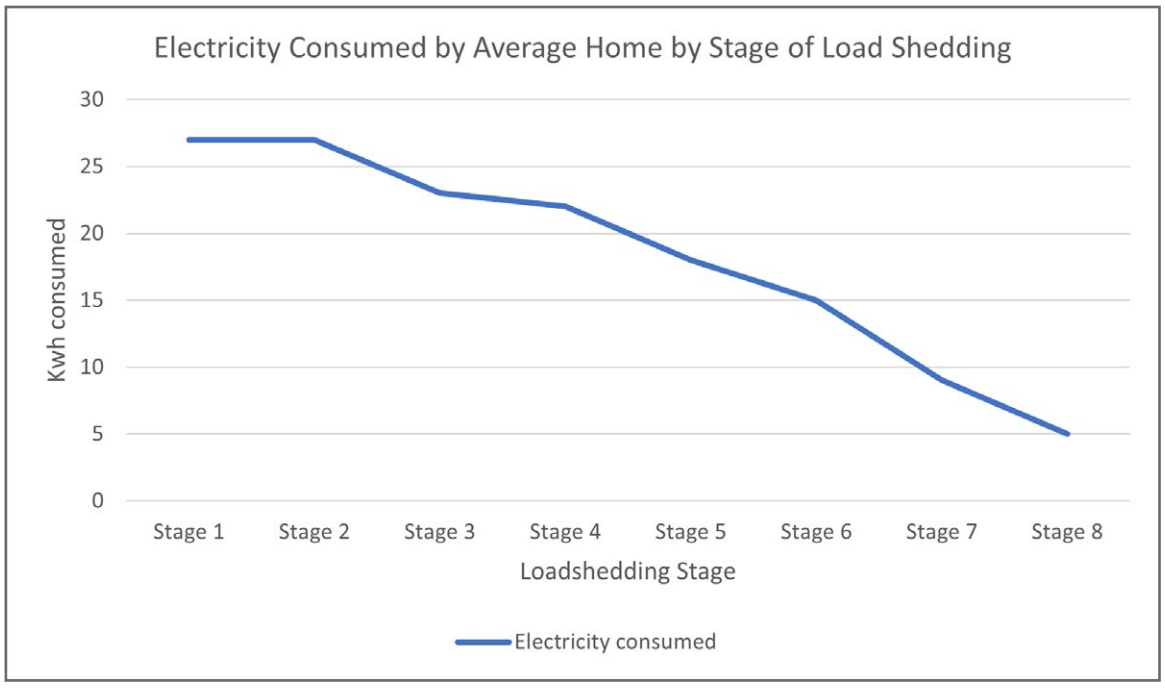
b) What kind of quadrilateral is ABCD? (1)

c) Why do you say so? (3)

(19)

Question 5 (12 marks)

1. In the graph below, the number of hours spent in load shedding vs the number of kilowatt hours of electricity consumed in an average home per day is shown. Study the graph carefully before answering the questions that follow.



a) Is the graph increasing or decreasing? (1)

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b) Which part of the graph is constant? (1)

c) Give the minimum average number of kilowatt hours used daily. (1)

d) Is the data in this graph discrete or continuous? (1)

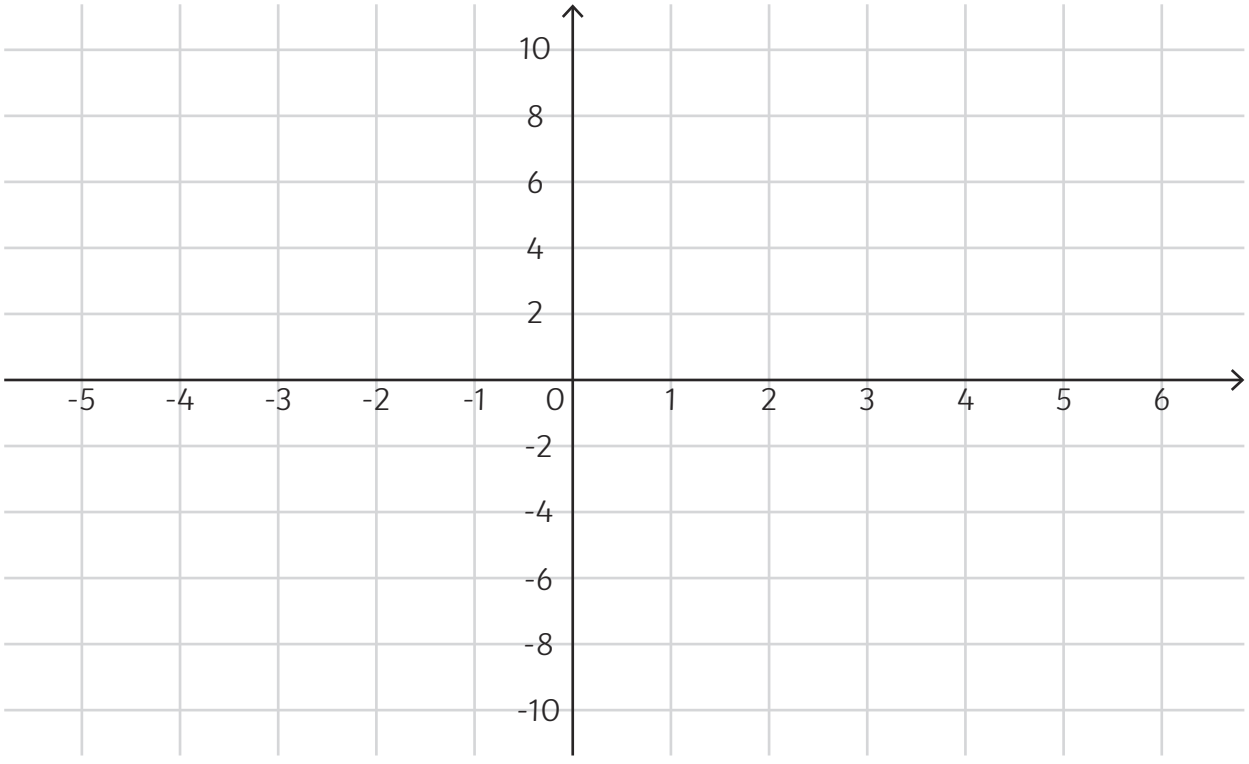
2. Answer the questions that follow based on the equation given:

$y = -2x + 3$

a) Complete the table of ordered pairs: (3)

x	-2	-1	0	1	2	
y						-15

b) Draw the graph using the ordered pairs on the Cartesian plane below: (4)



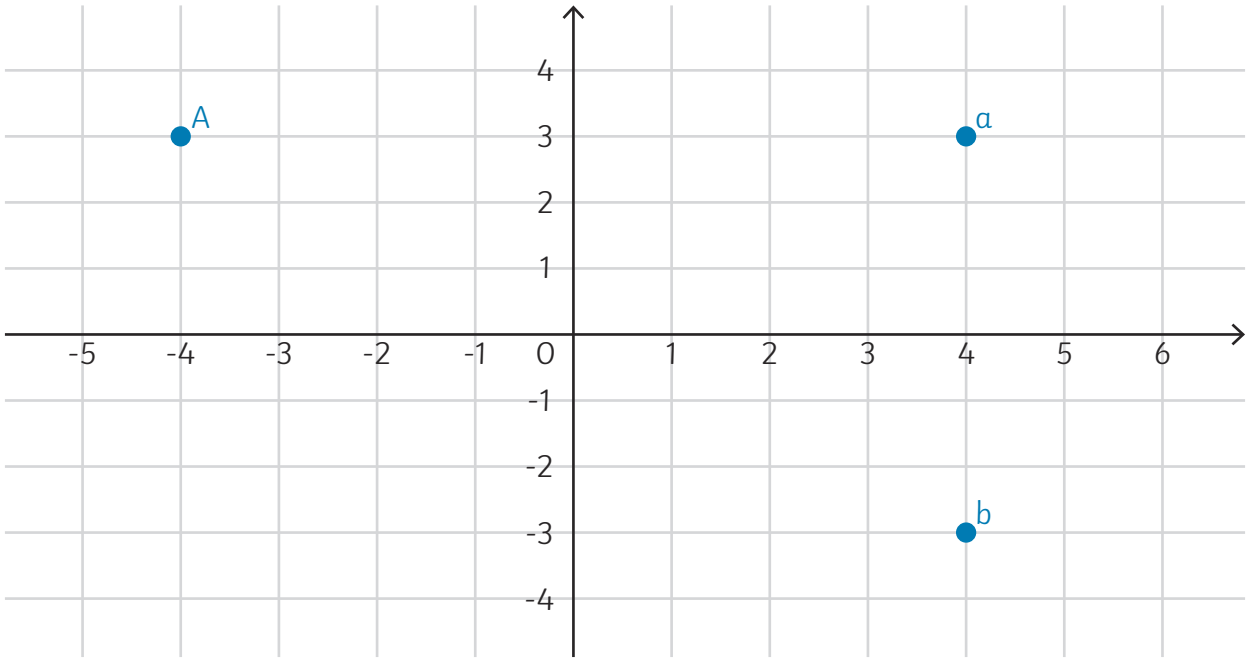
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c) What is the trend of the graph? (1)

(12)

Question 6 (7 marks)

1. Look carefully at the points below then answer the questions that follow:



a) What kind of transformation is A to a? (1)

b) What kind of transformation is A to b? (1)

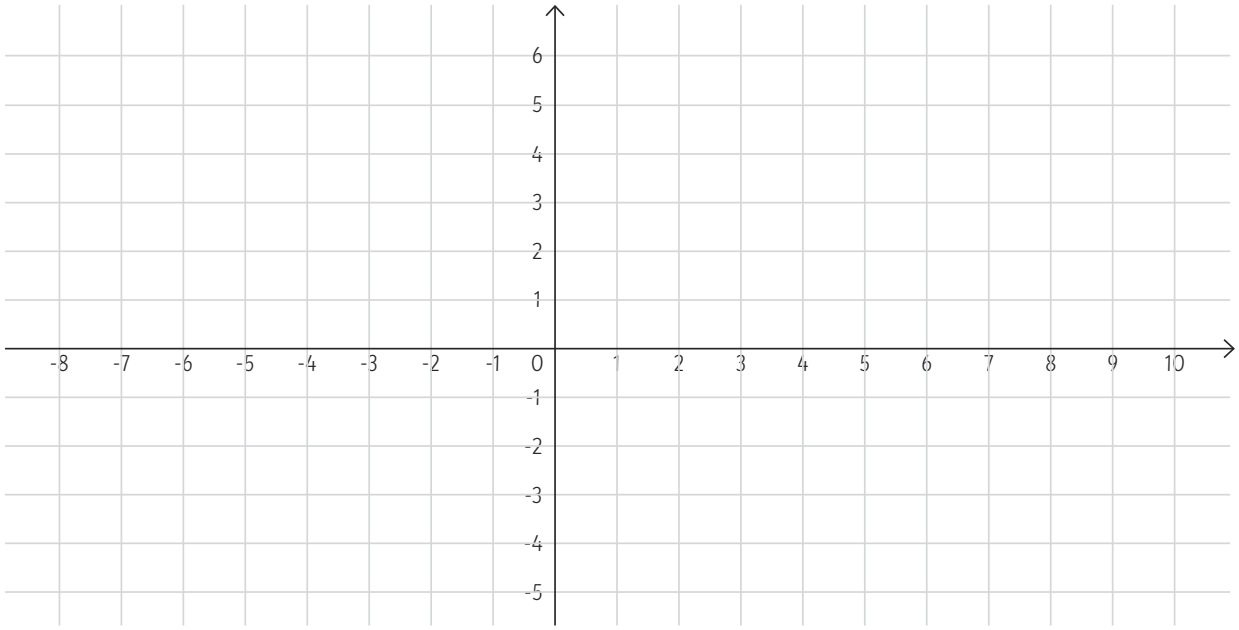
c) What kind of transformation is a to b? (1)

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2. Draw each of the following transformations onto the Cartesian plane given below:

   a) A (2, 3) translated 3 spaces to the left, and 4 spaces down. (2)

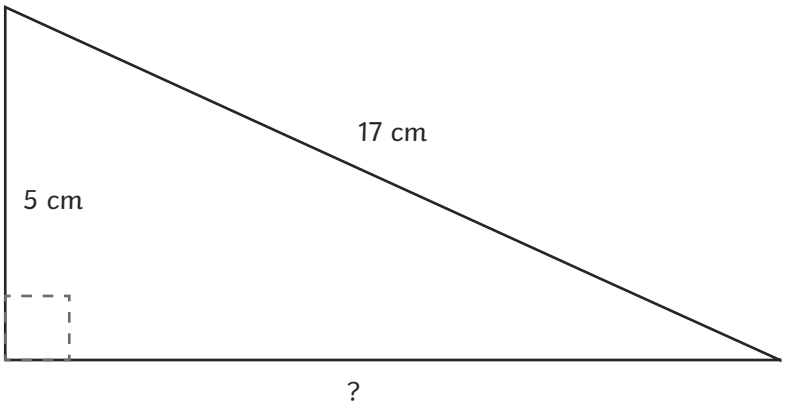
   b) B (-1, 5) reflected around the y-axis (2)



(7)

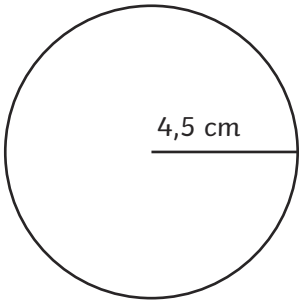
Question 7 (22 marks)

1. Find the value of the missing side: (4)



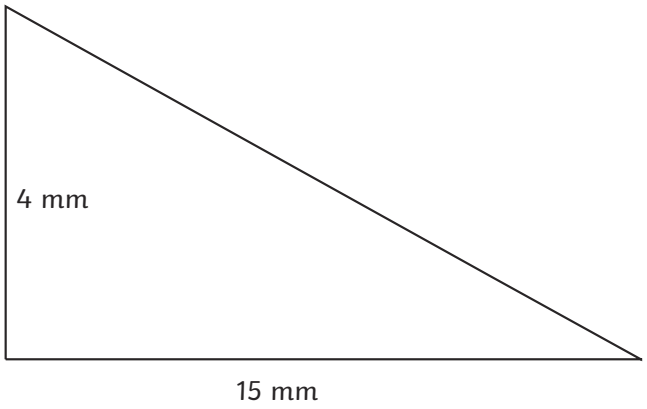
2. Find the area and the perimeter for each of these shapes:

   a) (4)

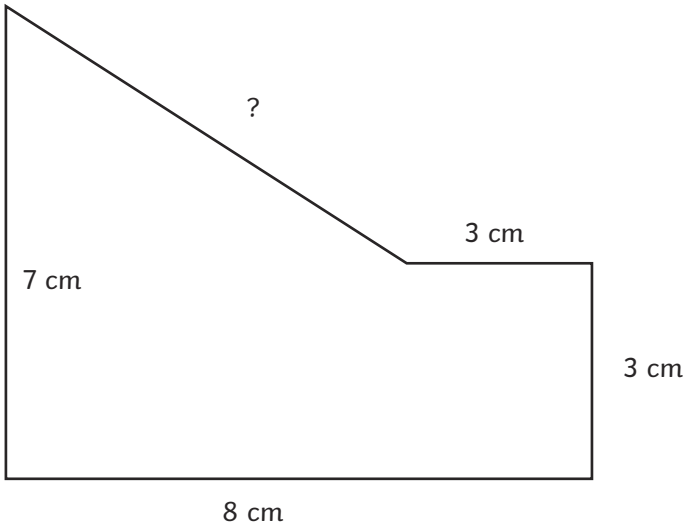


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b) (5)



c) (9)



(22)

Grand Total: 100

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Grade 8 Exam Paper November Answers ATP Based on Term 3 and Term 4 Work

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Instructions

1. Answer ALL the questions.
2. Write neatly and legibly
3. Use a ruler where necessary.
4. Read each question carefully.

Question 1 (12 marks)

1. Simplify the following expressions:

a) $3x(4x + 5) - 4x(7x - 8)$ (3)

$= 12x^2 + 15x - 28x^2 + 32x$ **(2 marks)**

$= -16x^2 + 47x$ **(1 mark)**

b) $\frac{27x^3 + 33x^2 - 12x}{3x} - 4x(5x - 8)$ (4)

$\frac{27x^3}{3x} + \frac{33x^2}{3x} - \frac{12x}{3x} - 20x^2 + 32x$ **(2 marks)**

$= 9x^2 + 11x - 4 - 20x^2 + 32x$ **(1 mark)**

$= -11x^2 + 43x - 4$ **(1 mark)**

c) $\sqrt{(81x^4)} + (-4x)^2$ (3)

$= 9x^2 + (16x^2)$ **(2 marks)**

$= 25x^2$ **(1 mark)**

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2. Find the value of the following expression if a = -1, b = 3, and c = 2 (2)

$ab^2 + bc - abc$

$= (-1)(3)^2 + (3)(2) - (-1)(3)(2)$ **(1 mark)**
$= -9 + 6 + 6$ **(1/2 mark)**
$= 3$ **(1/2 mark)**

(12)

Question 2 (14 marks)

1. Complete this table of ordered pairs for the equation $y = 3x - 4$ **(1 mark each)** (5)

x	-3	0	3	5	**8**
y	**-13**	**-4**	**5**	**11**	20

2. Find the value of x in each of these equations:

a) $7x + 2 = 23$ (2)

$7x = 23 - 2$

$7x = 21$ **(1 mark)**

$\frac{7x}{7} = \frac{21}{7}$

$x = 3$ **(1 mark)**

b) $3x^2 - 5 = 70$ (3)

$3x^2 = 70 + 5$

$3x^2 = 75$ **(1 mark)**

$x^2 = 25$ **(1 mark)**

$x = \sqrt{25}$

$x = \pm 5$ **(1 mark)**

c) $\frac{x}{3} + 4 = 2(x - 7)$ (4)

$\frac{x}{3} + 4 = 2x - 14$ **(1 mark)**

$4 + 14 = 2x - \frac{1}{3}x$ **(1 mark)**

$18 = 1\frac{2}{3}x$

$54 = 5x$ **(1 mark)**

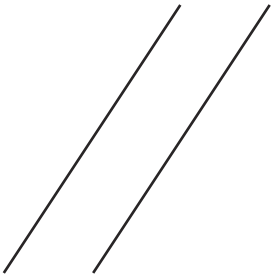
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$x = \frac{54}{5}$ or $10\frac{4}{5}$ **(1 mark)**

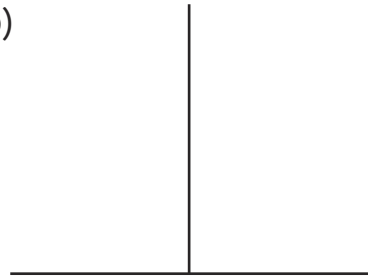
(14)

Question 3 (14 marks)

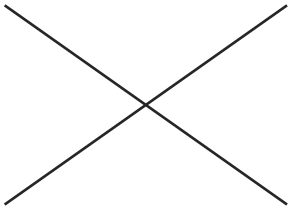
1. Name each of these types of lines: (3)

a) 

parallel

b) 

perpendicular

c) 

intersecting

2. Solve for the variables x, y, m and n in this diagram:
   In the diagram below,

   AB || CD, and GF ⊥ AB, $H\hat{D}F = 22°$, and $B\hat{H}D = 46°$

Statement	Reason
$x = 46°$	Vertically opp. ∠'s equal
$x + y = 180°$	Angles on a straight line = 180°
$y = 180° - 46°$	
$y = 134°$	
$m = 22°$	Alt ∠'s equal, AB || CD
$m + y + n = 180°$	Sum of ∠'s in Δ = 180°
$n = 180° - 22° - 134°$	
$n = 24°$	

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Question 4 (19 marks)

1. Match the shape in column A with the description given in column B. (7)

   1. **e** 2. **g** 3. **a** 4. **b** 5. **f** 6. **c** 7. **d**

2. How do we know when two shapes are congruent? (2)

   Two shapes are congruent when all the sides are equal and the angles are equal.

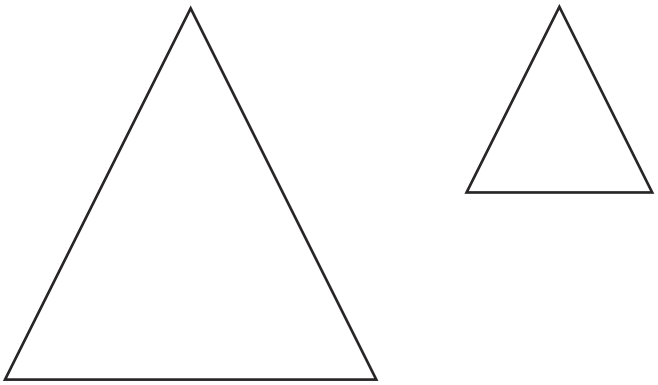
3. State whether each pair of shapes is similar, congruent, or neither. (3)

   a)



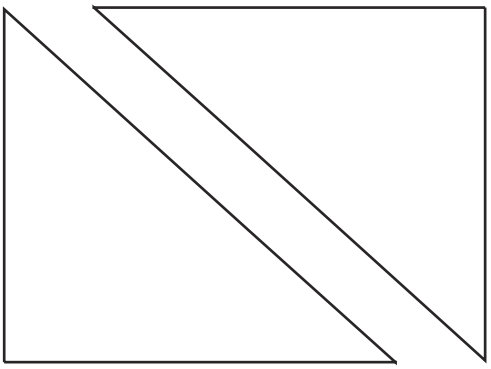
neither

   b)



similar

   c)



congruent

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4. In the diagram, AB || DC and AD || BC.
We are given that DE = EB = 3cm and AE = EC = 4cm.

a) Name three sets of congruent triangles: (3)

Any of these three:
ΔDEA = ΔBEC,
ΔADC = ΔABC,
ΔDEC = ΔAEB,
ΔDBA = ΔDCB

b) What kind of quadrilateral is ABCD? (1)

parallelogram

c) Why do you say so? (3)

both sets of opposite sides are parallel and diagonals bisect,
due to congruency, both sets of opposite sides are equal.

(19)

Question 5 (12 marks)

1. In the graph below, the number of hours spent in load shedding vs the number of kilowatt hours of electricity consumed in an average home per day is shown. Study the graph carefully before answering the questions that follow.

a) Is the graph increasing or decreasing? (1)

decreasing

b) Which part of the graph is constant? (1)

it is between stage 1 and stage 2

c) Give the minimum average number of kilowatt hours used daily. (1)

5 kwh

d) Is the data in this graph discrete or continuous? (1)

continuous

2. Answer the questions that follow based on the equation given:

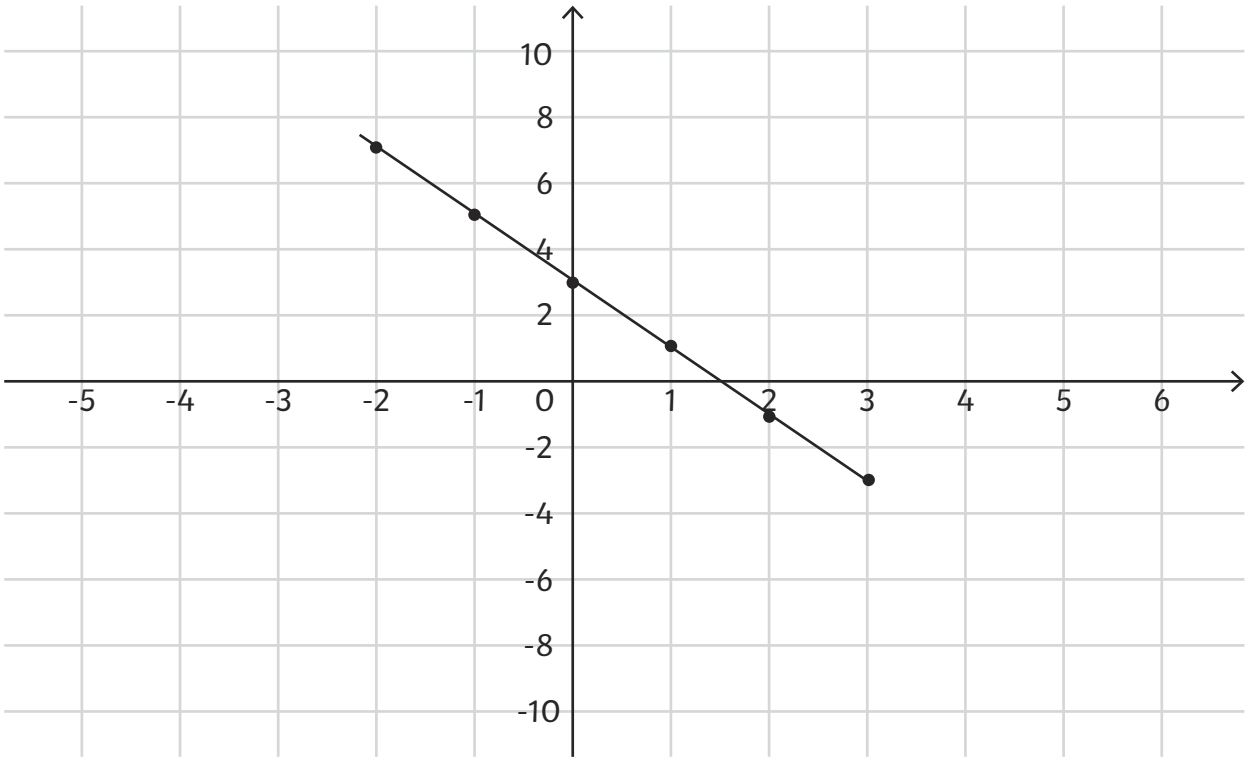
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$y = -2x + 3$

a) Complete the table of ordered pairs: **(1/2 mark each)** (3)

x	-2	-1	0	1	2	**9**
y	**7**	**5**	**3**	**1**	**-1**	-15

b) Draw the graph using the ordered pairs on the Cartesian plane below: (4)



c) What is the trend of the graph? (1)

The graph is decreasing. It has a negative trend.

(12)

Question 6 (7 marks)

1. Look carefully at the points below then answer the questions that follow:

a) What kind of transformation is A to a? (1)

reflected about the *y*-axis

b) What kind of transformation is A to b? (1)

translation

c) What kind of transformation is a to b? (1)

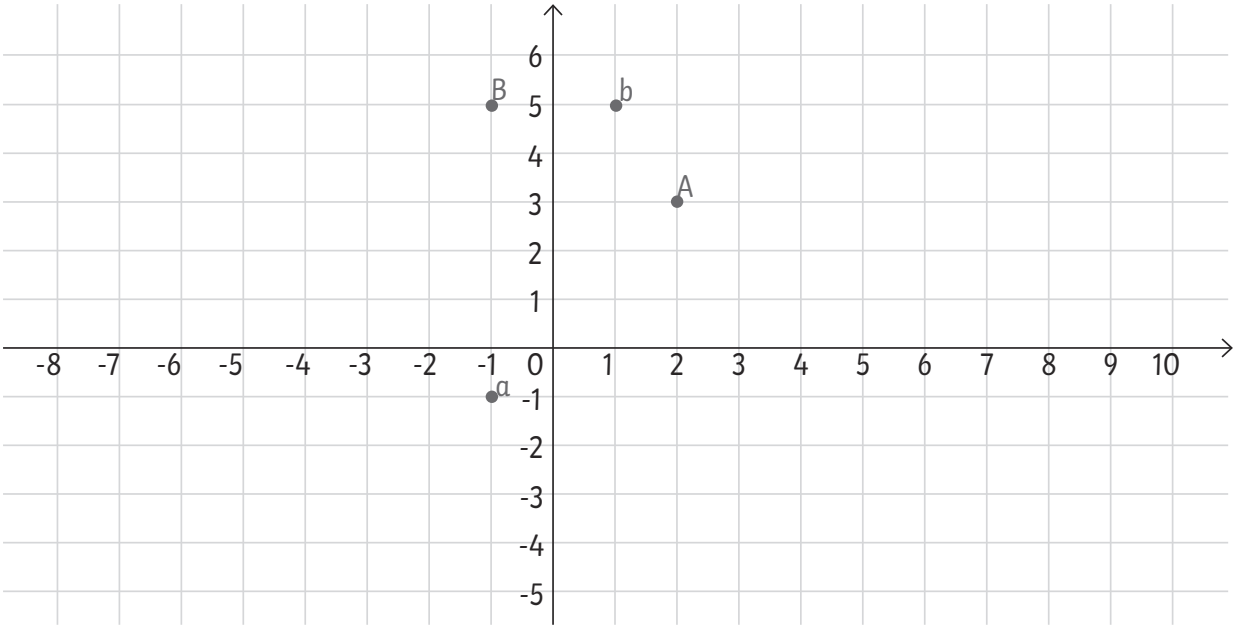
reflected about the x-axis

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2. Draw each of the following transformations onto the Cartesian plane given below:

a) A (2, 3) translated 3 spaces to the left, and 4 spaces down. (2)

b) B (-1, 5) reflected around the y-axis (2)



(7)

Question 7 (22 marks)

1. Find the value of the missing side: (4)

$h^2 = a^2 + b^2$

$(17\ cm)^2 = (5\ cm) + b^2$

$b^2 = 289\ cm^2 - 25\ cm^2$

$b^2 = 264\ cm^2$

$b = 2\sqrt{(66)}$ cm or 16,25 cm

2. Find the area and the perimeter for each of these shapes:

a) **Area = πr^2** **(4)**

Area = $\pi\ (4{,}5cm)^2$

Area = 63,62 cm^2

Perimeter = $2\pi r$

Perimeter = $2 \times \pi \times 4{,}5cm$

Perimeter = 28,27 cm

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b) **Area = $\frac{1}{2}b \times \perp h$** **(5)**

Area = $\frac{1}{2}$ x 15 mm x 4 mm

Area = 30 mm^2

Perimeter = side 1 + side 2 + side 3

Perimeter = 4 mm + 15 mm + 15,52 mm

Perimeter = 34,52 mm

$h^2 = a^2 + b^2$

$h^2 = (4\text{ mm})^2 + (15\text{ mm})^2$

$h^2 = 16\text{ mm}^2 + 225\text{ mm}^2$

$h^2 = 241\text{ mm}^2$

h = 15,52 mm

c) **Area = Area of triangle + area of rectangle** (9)

Area = $\frac{1}{2}$ b × ⊥h + l × b

Area = $\frac{1}{2}$ × 5 cm × 4 cm + 8 cm × 3 cm

Area = 10 cm^2 + 24 cm^2

Area = 34 cm^2

$h^2 = a^2 + b^2$

$h^2 = (7\text{ cm})^2 + (5\text{ cm})^2$

$h^2 = 49\text{ cm}^2 + 25\text{ cm}^2$

$h^2 = 74\text{ cm}^2$

h = 8,6 cm

Perimeter = 7 cm + 8 cm + 3 cm + 3 cm + 8,6 cm

Perimeter = 29,6 cm

(22)

Grand Total: 100

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